Hard AbsTopK gate on MNIST: after a linear layer against on the input

2026-10-26

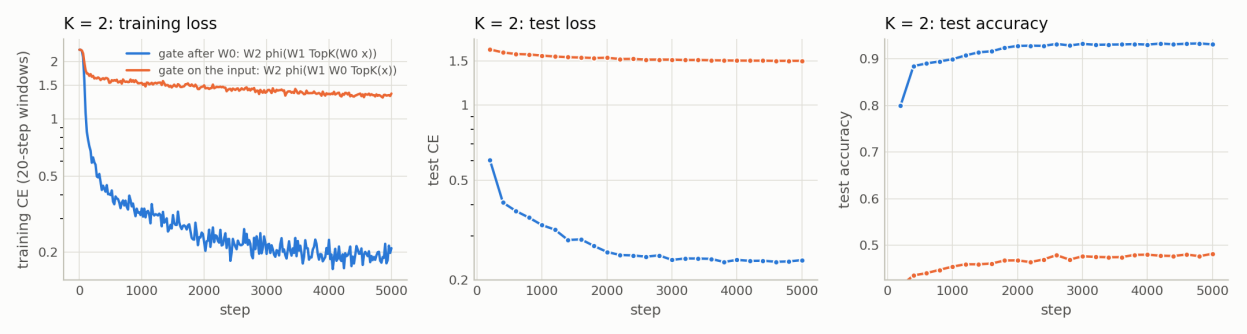


Figure 1: $K = 2$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for $z = W_2\,\phi(W_1\,\mathrm{AbsTopK}_K(W_0 x))$ (gate after W_0) and $z = W_2\,\phi(W_1 W_0\,\mathrm{AbsTopK}_K(x))$ (gate on the input); W_0 is 784×784 without bias, hidden width 1024, hard backward, 5000 steps.

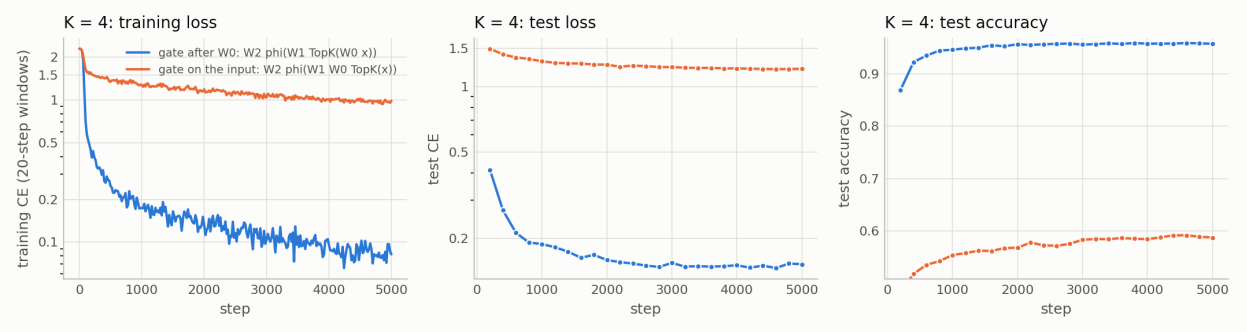


Figure 2: $K = 4$: as Figure 1.

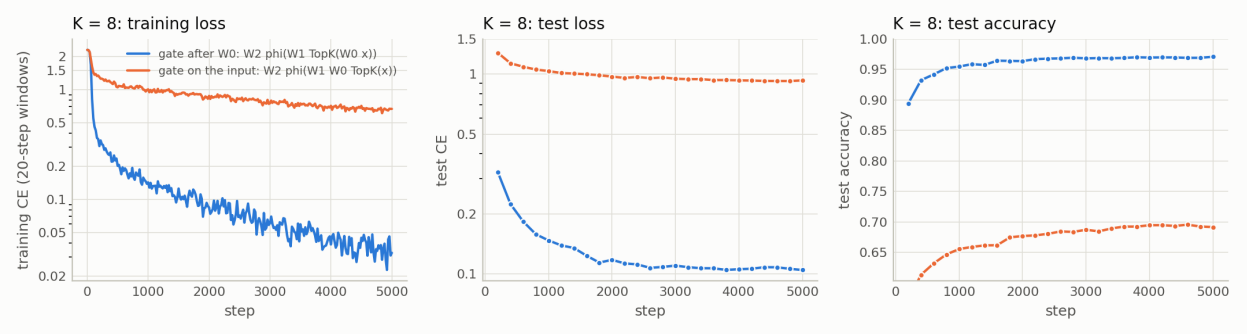


Figure 3: $K = 8$: as Figure 1.

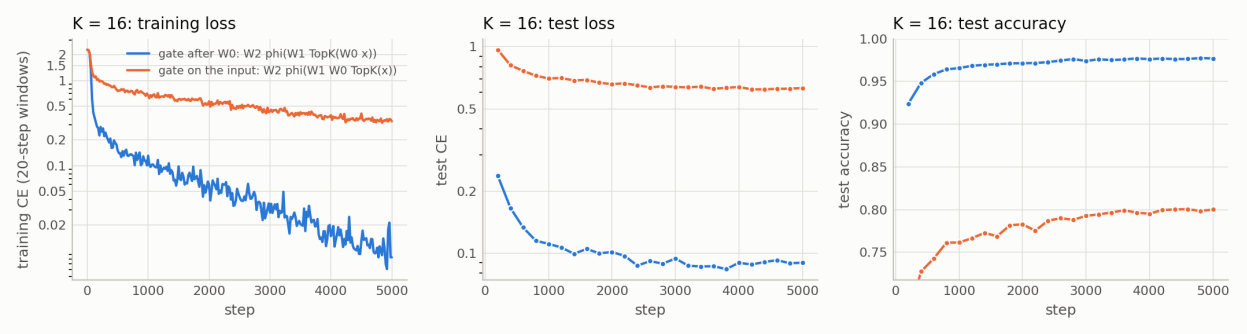


Figure 4: $K = 16$: as Figure 1.